	TOTAL FOR QUESTION 2	**8**
3(a)	Electric potential at a point is the work done per unit charge by an external force in bringing a small positive test charge from infinity to that point.	**B1**
3(b)(i)	$E = -\frac{dV}{dr}$ $\lvert E \rvert$ = gradient of graph $= \left\lvert \frac{2.50 - 0}{(0 - 2.50) \times 10^{-2}} \right\rvert$ [M1] $= 100 \text{ N C}^{-1}$ [A1] Acceptable range: 88 <=E<= 1105□□ Direction: in positive x-direction / towards right / from left to right / → [B1]	**M1** **A1** **B1**
3(b)(ii)	The graph is symmetrical.	**B1**
3(b)(iii)	Using a point such as x = 0.20 cm where V ≈ 6.0 V: $V = \frac{Q}{4\pi\varepsilon_o x} + \frac{Q}{4\pi\varepsilon_o (0.040 - x)}$ $6.0 = \frac{Q}{4\pi\varepsilon_o}\left(\frac{1}{0.002} + \frac{1}{0.040 - 0.002}\right)$ [M1] $Q = 1.27 \times 10^{-12} \text{C}$ [A1] Accept also if student used other suitable points for substitution.	**M1** **A1**
	TOTAL FOR QUESTION 3	**7**
4(a)	Resistance of a circuit component is the ratio of the potential difference across the	**B1**